Q1: Chatbot-Based Banking Assistant

A bank provides a chatbot for:

- Balance enquiry
- Transaction alerts
- Loan status

Tasks:

- Design using:
  - Amazon Lex
  - AWS Lambda
  - Amazon SNS
  - IAM
- Explain:
  - Intents, slots, backend flow
  - How notifications are triggered
  - Security using IAM roles

Q2: Secure Deployment + Chatbot Integration

The same platform wants:

- Quick deployment of backend
- A chatbot for order tracking

Tasks:

- Use:
  - AWS Elastic Beanstalk
  - Amazon Lex
  - IAM
- Explain:
  - How Beanstalk simplifies deployment
  - How Lex integrates with Lambda backend
  - IAM role design for chatbot
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Q3: Platform Deployment & Scaling

Platform must:

- Be deployed quickly
- Auto-scale during exams
- Maintain uptime

Tasks:

- Use:
  - AWS Elastic Beanstalk
  - Elastic Load Balancing
  - Auto Scaling Groups
- Explain:
  - Deployment workflow
  - Scaling triggers
  - Multi-AZ concept

Q4: End-to-End Cloud Architecture

Design a system where:

- Users interact via web + chatbot
- Requests processed asynchronously
- Notifications sent
- System scales dynamically

Must include ALL:

- Amazon Lex
- AWS Lambda
- Amazon SQS
- Amazon SNS
- Elastic Load Balancing
- Auto Scaling Groups
- AWS Elastic Beanstalk
- IAM

Tasks:

- Draw architecture
- Explain complete data flow
- Identify failure points

Q5: Booking Workflow

When a user books a ticket:

- Request handled by web app
- Booking processed asynchronously
- Notifications sent

Tasks:

- Use ALL:
  - Elastic Load Balancing
  - Auto Scaling Groups
  - Amazon SQS
  - AWS Lambda